



Diagnosis of heart disease using oversampling methods and decision tree classifier in cardiology

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Abstract

Purpose Heart disease is one of the most prevalent and critical diseases that endangers the lives of human beings. In addition to clinical diagnosis, machine learning and deep learning-based approaches are vital in the diagnosis of heart disease.

Method This paper proposes a balanced and optimized machine-learning algorithm for heart disease detection. This technique combines oversampling techniques, attribute pruning, CART decision tree classifier, and rule pruning through hyperparameter tuning to identify the presence of heart disease. It further identifies the key attributes that contribute to the occurrence of heart malfunctioning.

Results Experimental results show that SMOTE sampled dataset exhibits effective performance when implemented using a balanced and optimized machine learning algorithm, with an improvement of 11%, 75%, 62%, and 71% in accuracy, precision, recall, and f1 scores when compared with the dataset that was not subjected to sampling. The algorithm works effectively when the imbalance ratio is high for a dataset.

Conclusion The algorithm can be used to predict the presence of heart disease even in highly imbalanced datasets and identify critical features in the malfunctioning of the heart.

Keywords BOML · heart disease · machine learning · oversampling · decision tree

Introduction

The primary cause of uncertainty in life is health-related issues. The COVID-19 scenario emphasizes the need for modern technology that will strongly bind with the traditional system of medical diagnosis and treatment, to realize

a wholesome technology-supported expert medical system. Prevention is always better than cure. Prevention of diseases requires a detailed analysis of the root causes of the conditions. Statistical analysis of factors that contribute to diseases is very critical due to the high level of correlation between the factors. An enormous amount of data is available for analysis. However, efficient data mining techniques have to be applied for handling the data and making them usable for arriving at efficient decisions.

Many of the techniques available to mine relevant data are black boxes in nature. An efficient technique should be transparent and exhibit efficient decision-making to analyze data related to the root cause of diseases. Eventually, the decision tree is the most familiar technique that generates a series of rules to arrive at a decision (Navada et al. 2011). A decision tree is a decision support tool that uses a tree-like model of decisions and possible consequences, including chance event outcomes, resource costs, and utility. Decision trees are commonly used in operations research, specifically in decision analysis, to help identify a strategy most likely to reach a goal, but they are also a popular tool in machine learning. It is a predictive modeling tool that constructs decision trees via an algorithmic approach

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that identifies ways to split a data set based on different conditions. They are a non-parametric supervised learning method used for both classification and regression tasks. The goal is to create a model that predicts the value of a target variable by learning simple decision rules inferred from the data features.

The majority of the medical datasets available for data analysis is imbalanced. Imbalanced datasets refer to the mismatch in the number of majority and minority classes. The imbalanced nature becomes the prime difficulty in decision-making as there are fewer samples in one class when compared to another. This paper presents a balanced and optimized machine learning (BOML) model. This technique applies oversampling techniques such as synthetic minority oversampling technique (SMOTE) (Chawla et al. 2018), ADASYN (Haibo et al. 2008), borderline-SMOTE (Hui et al. 2005), safe-level SMOTE (Bunkhumpornpat et al. 2009), and Rose (Lunardon et al. 2014) on Statlog (heart) datasets that are imbalanced in nature. Consequently, the decision tree algorithm is applied on the sampled heart datasets. This facilitates the decision tree algorithm to treat both classes with equality and aids in arriving at transparent rules, needed for arriving at conclusions. Attribute pruning is applied on the developed model to identify the key attributes that contribute to the risk of the disease. Rule pruning is done by hyper-parameter tuning using tenfold grid search cross-validation. Best parameters are extracted and applied on the decision tree classifier, to make efficient predictions. The proposed model is validated using a tenfold cross-validation technique, applied over medical datasets, extracted from the UCI and BioLNCC repositories.

The paper is composed of three sections. Section II analyses the widespread use of decision trees in medical expert systems. Section III details the materials and the detailed experimentation involved in the development of the proposed BOML model. Section IV presents the experimental results. Section “Discussion” gives a brief discussion on the experimental results. Section “Conclusion” concludes the paper.

Literature survey

Medical expert systems blended with medical diagnosis aid in self-prediction of the occurrence of a disease. This innovative approach uses various machine learning algorithms like ANN (artificial neural networks), SVM (support vector machine), and decision tree to assist the experts in predicting correctly. Owing to the transparency and distinct understanding offered by the varied range of attributes, the decision tree algorithm was predominantly used for classification and regression problems (Ghiasi 2020) (Ahsan et al. 2022). However, decision trees face the challenges of overfitting, parameter tuning, and biased learned trees.

Lately, many exciting works related to the widespread usage of decision trees in medical diagnosis have been gaining importance. To predict the possibility of Alzheimer’s disease in patients, Dana et al. (D. AL-Dlaeen and A. Alashqur 2014) used a decision tree that adopted induction-based approach. The Alzheimer’s disease model performed prediction using major attributes such as gender, age, genetic reasoning, brain injury, and vascular disease. Entropy or information gain was iteratively computed for each attribute, eventually building the nodes of the decision tree. However, to increase the precision of detection, the decision tree has to be periodically refreshed by updating the data of the cases whose disease conditions have been known. An improvised C4.5 decision tree method utilizing RELIEFF attribute weighting techniques for medical diagnosis was proposed by Liu et al. (Liu et al. 2016). This model used the C4.5 decision tree and removed the irrelevant attributes using the RELIEFF algorithm to build an accurate distributed model for medical diagnosis. Though the accuracy rate of prediction was improved, the computation time in building and prediction was increased. Albu (Albu 2017) used a decision tree for hepatitis prediction. Decision trees were used for predictions regarding hepatitis infection, and the conclusion was drawn exclusively based on hepatitis markers. Pashaei et al. (Pashaei et al. 2015) used a hybrid decision tree approach using boosted C5.0 decision tree and a C4.5 decision tree. C4.5 decision tree generated a unique function to improve the performance of particle swarm optimization (PSO). The C4.5 decision tree was deployed for medical diagnosis. Stasis et al. (Stasis et al. 2003) used data mining/distillation methods to analyze heart sounds recorded from patients affected with aortic stenosis (AS) or mitral regurgitation (MR). Amiri et al. (Amiri and Armano 2013) used classification and regression trees (CART) (Breiman 1984) to develop a diagnostic system for heart disease classification and detection. J.P.Li et al. (Sivaranjani et al. 2019) have used feature selection algorithms to predict the presence of heart disease. R. Sivaranjani et al. (“A hybrid intelligent system framework for the prediction of heart disease using machine learning algorithms” xxxx) have proposed a genetic algorithm-based decision tree for disease classification. Amin Ul Haq (Li et al. 2020) et al. had proposed a hybrid approach where various classifiers were combined with feature selection algorithms to predict the disease. The diagnostic system designed and implemented for detecting and classifying heart diseases assists physicians in the auscultation of patients, by reducing the number of unnecessary echocardiograms. It is observed that decision trees are widely used in medical diagnosis across varied dimensions.

Medical diagnosis has been witnessing rapid developments in the past decades. However, efficient data handling techniques are the prime requirement to make algorithms successful in clinical diagnostic systems (Ghiasi 2020)

(Ahsan et al. 2022). A fundamental issue in data is class imbalance. Recent research works have been predominant in working on the class imbalance problem (Rath et al. 2022). Though sampling techniques such as synthetic minority oversampling technique (SMOTE) (Chawla et al. 2018), ADASYN (), borderline-SMOTE (Hui et al. 2005), safe-level SMOTE (Bunkhumpornpat et al. 2009), and Rose (Lunardon et al. 2014) are prevalent, the model developed for diagnosis should be immune to the problem of overfitting. The proposed BOML model in this paper has overcome the problem of overfitting.

This paper focuses on a hybrid approach where oversampling techniques coupled with attribute pruning, rule pruning, and hyper-parameter tuning implemented over decision tree classifiers are applied on imbalanced Framingham Heart Study (FHS) (<https://biolincc.nhlbi.nih.gov/studies/framesort/>, . xxxx) and Statlog datasets (Dua and Graff 2019).

Materials and experiments

An experiment was performed to evaluate the efficiency of conventional and proposed methods. This section deals with the software tools used for implementation, the dataset for investigation, the selection of training and testing datasets, and the validation of the network.

Dataset description

The experiment is performed on the FHS (FHS 2015) (<https://biolincc.nhlbi.nih.gov/studies/framesort/>) and Statlog heart dataset (Dua and Graff 2019) as depicted in Table 1.

Framingham's dataset comprising epidemiologic CVD data was obtained from the Biologic Specimen and Data Repository Information Coordinating Center (BioLINCC) of the US National Institutes of Health (NIH). The FHS dataset contains 4240 observations with 15 attributes. The Statlog heart dataset contains 270 observations with 13 attributes. Data pre-processing was not required as there was no missing data.

Training and testing datasets

The training and testing data split were chosen randomly in the ratio of 9:1. The training and testing samples for Framingham and Statlog heart datasets are depicted in Table 1. tenfold cross-validation is applied to the datasets to train, validate, and test the data.

Need for oversampling algorithms

The FHS dataset contains a total of 4240 samples which evaluate to 3658 samples after the removal of NaN values in the dataset. A total of 3101 samples and 557 samples belong to class 0 and class 1, respectively. Class 0 and class 1 represent the absence and presence of 10 years of coronary heart disease (CHD) risk in patients respectively with a class imbalance ratio of 1:6. Statlog dataset contains 270 samples out of which 150 samples belong to class 1 and 120 samples belong to class 2. The data depicts two classes—absence of heart disease (2) and presence of heart disease (1) with a class imbalance ratio of 1.25:1. This illustrates that the FHS dataset is highly imbalanced when compared to Statlog. Before fitting the samples with the decision tree, sampling algorithms are applied to the datasets. This prevents the creation of biased decision trees based on classes that dominate. SMOTE, borderline-SMOTE, ADASYN, rose, and safe level SMOTE oversampling algorithms handle the class imbalance in the dataset.

Decision tree classifier

The objective of the model to be developed is to predict the presence or absence of heart disease by learning decision rules based on the features described by the data. In this paper, the CART algorithm is applied to the sampled datasets. CART uses Gini's impurity index as the criterion to build decision trees for classification. The algorithm works as depicted in Fig. 1.

Table 1 Datasets

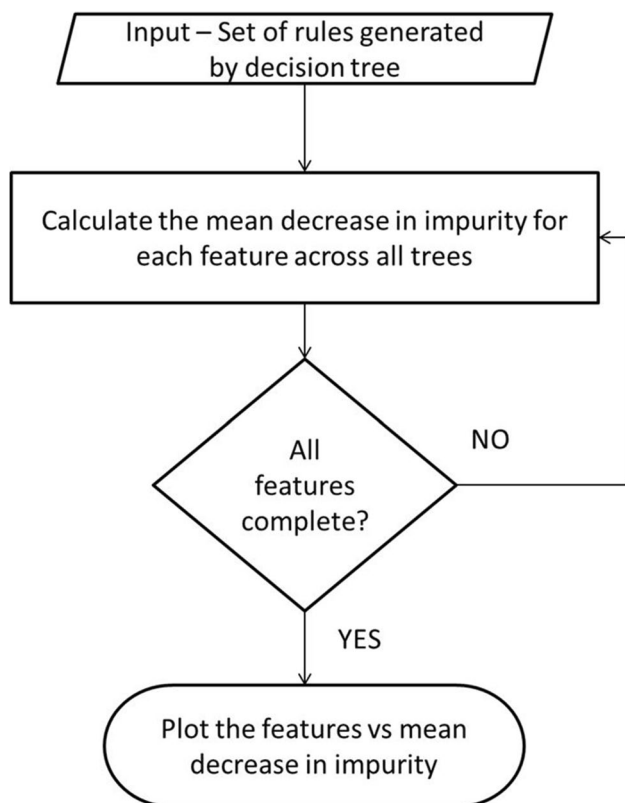
Dataset	Attributes	Data count			Data split-up count			
		Total	Presence	Absence	Training data		Test data	
					Presence	Absence	Presence	Absence
FHS[]	15	4240	644	3596	495	2797	62	304
Statlog[]	13	270	150	120	134	109	16	11

Fig. 1 CART algorithm

//CART Algorithm//
Input: Features of dataset.
Output: Decision tree.
Step 1: Find best split for each feature
Step 2: Find the node's best split which maximizes the splitting criterion.
Step 3: Split the node using best node split from Step 2 and repeat from Step 1 until stopping criterion is satisfied.

Attribute pruning

The purpose of attribute pruning is to identify the significant attributes from the selected rule, based on the impurity of each attribute or feature. Traversing across all trees, the mean decrease in impurity for each feature is computed. Once the CART decision tree classifier model is fitted, it provides an attribute feature_importances_, which obtains the scores of each feature. The attribute pruning methodology adopted in the proposed BOML model is shown in Fig. 2.

**Fig. 2** Attribute pruning

Rule pruning

Pruning reduces the size of decision trees by removing redundant parts in the tree that are not sufficient to classify instances. Decision trees are highly subjected to overfitting. Effective pruning can be used to overcome overfitting issues. Pruning progressively removes error-prone components and improves the classifier's performance. In the rule pruning process, each rule is explicitly tested for its ability to correctly identify all the positive classes in a dataset. The rule that classifies the patterns error-free with the maximum count is selected. The methodology to perform rule pruning in the proposed BOML model is depicted in Fig. 3.

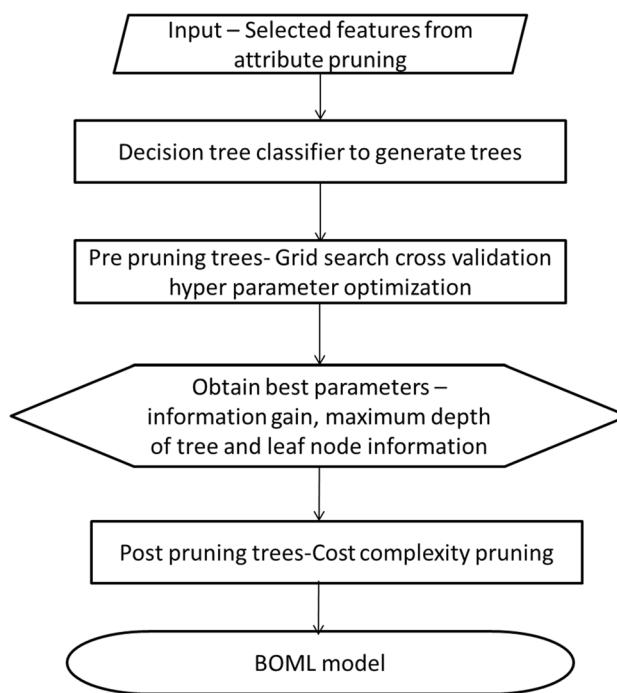
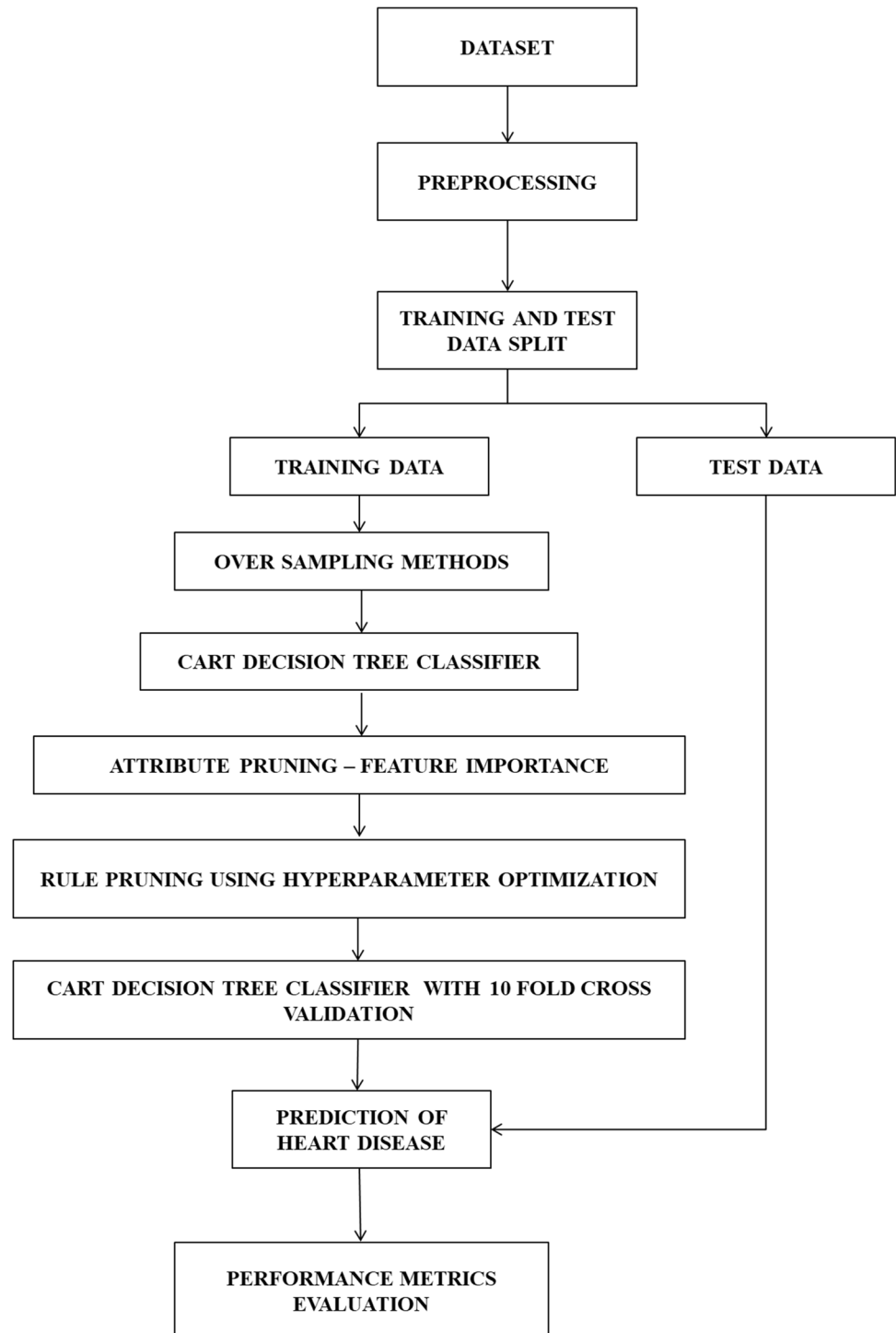
**Fig. 3** Rule pruning

Fig. 4 Flow methodology for BOML algorithm



Validation and prediction

The BOML model developed after the post-pruning stage is subjected to tenfold cross-validation training. The model is fitted and subjected to test data to predict the presence of disease.

Proposed balanced and optimized machine learning (BOML) for disease detection

This paper proposes a BOML model that oversamples the minority classes to balance the dataset, prunes the attributes to select the key features, learns it using CART decision trees, and performs hyper-parameter optimization and pruning of trees to predict the presence of disease. Algorithm 1 depicts the pseudocode of the proposed BOML algorithm. The methodology of the BOML algorithm is depicted in Fig. 4.

Algorithm 1 Pseudo-code of proposed BOML model

- 1: Begin
- 2: Pre-processing FHS and Statlog heart datasets
- 3: Apply oversampling algorithms on the datasets
- 4: Perform attribute pruning to select key features that influence the target column
- 5: Train the selected features of dataset with CART Decision tree classifier
- 6: Perform pre pruning of trees using hyper parameter optimization to obtain best parameters
- 7: Fit the best parameters on the model and perform post pruning to reduce tree complexity
- 8: The BOML model is cross validated using tenfold cross-validation
- 9: Final evaluation is done using test data
- 10: Calculate performance evaluation metrics
- 11: End

Experiment setup

The proposed model is implemented on the Anaconda platform—Jupyter notebook using Python. tenfold cross-validation is applied to the FHS and Statlog datasets. A single fold contains ten subsets each with 27 data samples. Out of the ten subsets, nine subsets are subjected for training and one subset for testing. The fold is repeated ten times. The performance metric reported by the tenfold

cross-validation technique is the average of the values computed in each loop. The original dataset and sampled datasets are independently subjected to the CART algorithm, followed by attribute pruning and rule pruning algorithms. The machine learning model parameters are again evaluated based on hyper-parameter tuning, to obtain the best fit.

Performance evaluation metrics

Different performance evaluation metrics are derived to determine the efficiency of disease detection and identification of root cause. Figure 5 illustrates the fundamental binary classification matrix essential for deriving the metrics.

Performance evaluation metrics for disease detection derived from the classification matrix are shown in Eqs. 1, 2, 3, 4, and 5.

$$\text{Balance ratio} = \frac{\text{Minority classes}}{\text{Majority classes}} \quad (1)$$

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)} \times 100 \quad (2)$$

$$\text{Recall} = \frac{TP}{(TP + FN)} \times 100 \quad (3)$$

$$\text{Precision} = \frac{TP}{(TP + FP)} \times 100 \quad (4)$$

$$F1 \text{ Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

The area under the curve of the receiver operator characteristic (AUC-ROC) metric is an overall performance metric of a machine learning model. Models with higher AUC refer to an optimistically good decision-building model.

Experimental results

Results of data pre-processing and oversampling methods

Figure 6 depicts the correlation among the attributes of the FHS dataset having 4240 instances with 15 attributes and one target variable. Figure 7 shows the correlation among the features of the Statlog heart dataset having 297 instances,

Fig. 5 Binary classification matrix

	Predicted healthy person (negative)	Predicted heart diseased person (positive)
Actual healthy person (negative)	TN(True Negative)	FP(False Positive)
Actual heart diseased person (positive)	FN(False Negative)	TP(True Positive)

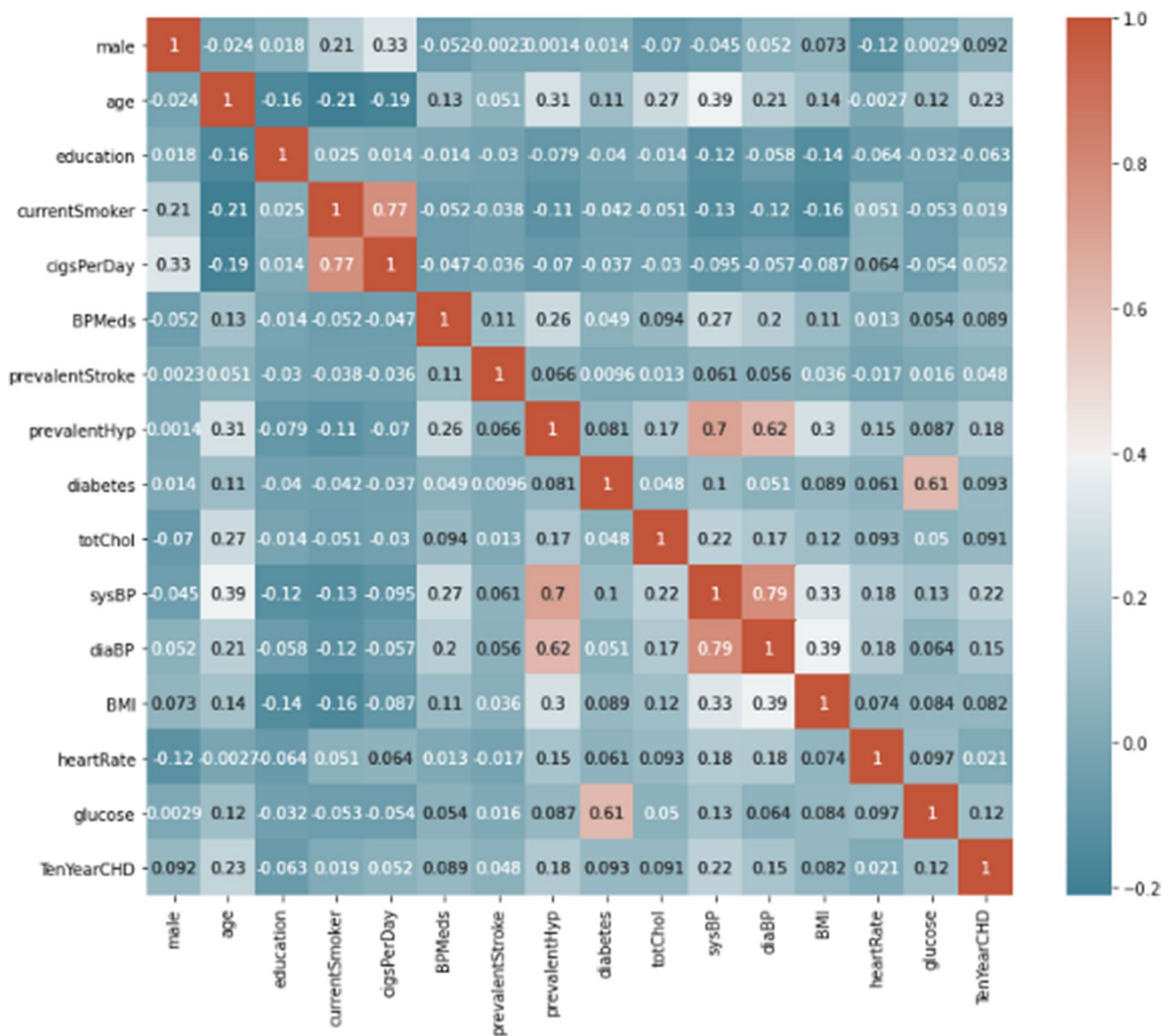


Fig. 6 Correlation matrix of the FHS heart dataset

13 attributes, and one target variable, generated using a heat map.

Table 2 presents the ratio of majority to minority samples in the original and oversampled datasets of FHS heart disease and Statlog heart datasets. The class imbalance ratio B/A of oversampled algorithms has improved significantly when compared to the original dataset.

Results of the BOML algorithm

The performance evaluation metrics—accuracy, precision, recall, f1 score, and AUC-ROC of the BOML algorithm for the test data of both datasets—are presented in Table 3.

The performance metrics of the original datasets, when validated on the CART decision tree classifier alone and

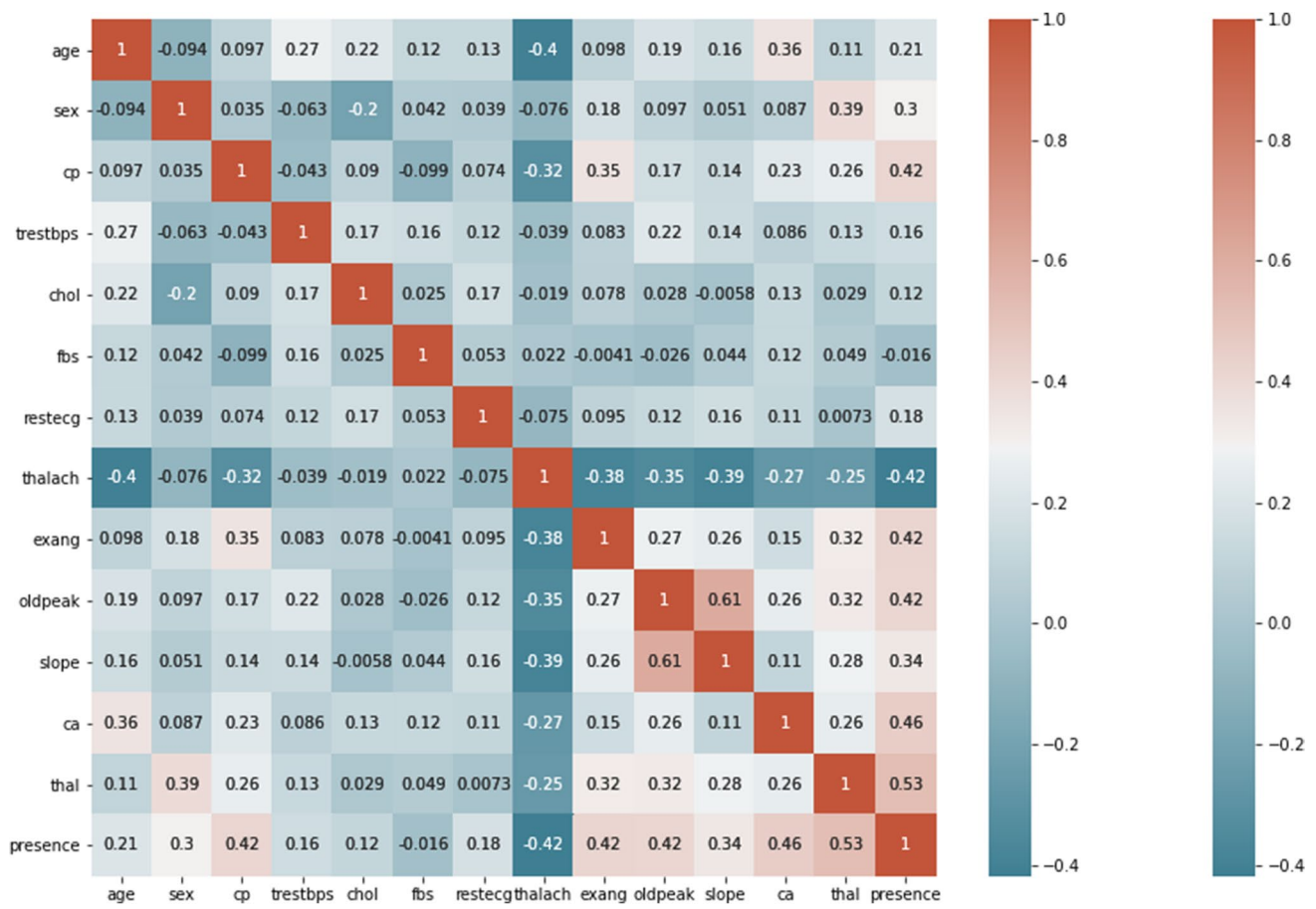


Fig. 7 Correlation matrix of Statlog heart dataset

Table 2 Original and oversampled datasets

	Total number of samples	Number of majority samples (A)	Number of minority samples (B)	B/A
Framingham heart study dataset				
Original dataset	4240	3596	644	5.58
SMOTE	11,188	5594	5594	1
Adasyn	6309	3208	3101	1.03
Borderline SMOTE	6202	3101	3101	1
Rose	6202	3101	3101	1
Safe level SMOTE	5234	3101	2133	1.45
Statlog heart dataset				
Original dataset	270	150	120	1.25
SMOTE	288	150	138	1.08
Adasyn	285	150	135	1.11
Borderline SMOTE	288	150	138	1.08
Rose	288	150	138	1.08
Safe level SMOTE	285	150	135	1.11

Table 3 Performance evaluation metrics of the BOML algorithm on FHS and Statlog heart datasets

Samples	Accuracy	Precision	Recall	f1 score	AUC ROC
Framingham heart study dataset					
Original—unsampled	0.713	0.198	0.316	0.232	0.55
SMOTE	0.801	0.79	0.832	0.807	0.68
Adasyn	0.777	0.786	0.788	0.78	0.70
Borderline SMOTE	0.781	0.764	0.821	0.787	0.65
Rose	0.665	0.677	0.64	0.652	0.58
Safe level SMOTE	0.572	0.524	0.534	0.526	0.53
Statlog heart dataset					
Original—unsampled	0.717	0.55	0.5	0.617	0.59
SMOTE	0.792	0.65	0.5	0.55	0.75
Adasyn	0.867	0.75	0.667	0.65	0.64
Borderline SMOTE	0.817	0.517	0.55	0.513	0.78
Rose	0.633	0.55	0.75	0.567	0.75
Safe level SMOTE	0.76	0.75	0.76	0.75	0.75

without being subjected to the BOML algorithm, are shown in Table 4 for comparison.

The AUC ROC curves of the BOML algorithm implemented on FHS and Statlog heart datasets are shown in Figs. 8 and 9, respectively.

Training and testing accuracies

The BOML algorithm is based on the fact that the datasets are subjected to oversampling algorithms. Hence it is essential that the model is not overfitting. An optimized value of the cost complexity alpha parameter is critical in creating a generalized decision tree. The training and test accuracies for various values of cost complexity alpha parameter for FHS and Statlog datasets are shown in Fig. 10 and Fig. 11 respectively.

Generated decision trees

The decision tree generated with the BOML algorithm applied on the FHS dataset with rose sampling is shown in Fig. 12. Rule pruning yielded an optimal decision tree with the best fit for the learning model, with a maximum depth of 3, as shown in Fig. 11.

Identification of the root cause of the disease

Attribute pruning identifies the significant attributes that are the root cause of the disease for SMOTE sampled FHS dataset. The significant attributes obtained using the feature importance attribute are shown in Fig. 13. Root cause identification aids in better treatment and prevention of disease.

Table 4 Performance evaluation metrics of the CART decision tree classifier alone without being subjected to BOML algorithm

Dataset subjected to CART classifier alone	Accuracy	Precision	Recall	f1 score	AUC ROC
Framingham heart study dataset					
	0.727	0.257	0.214	0.278	0.5
Statlog heart dataset					
	0.721	0.65	0.51	0.55	0.55

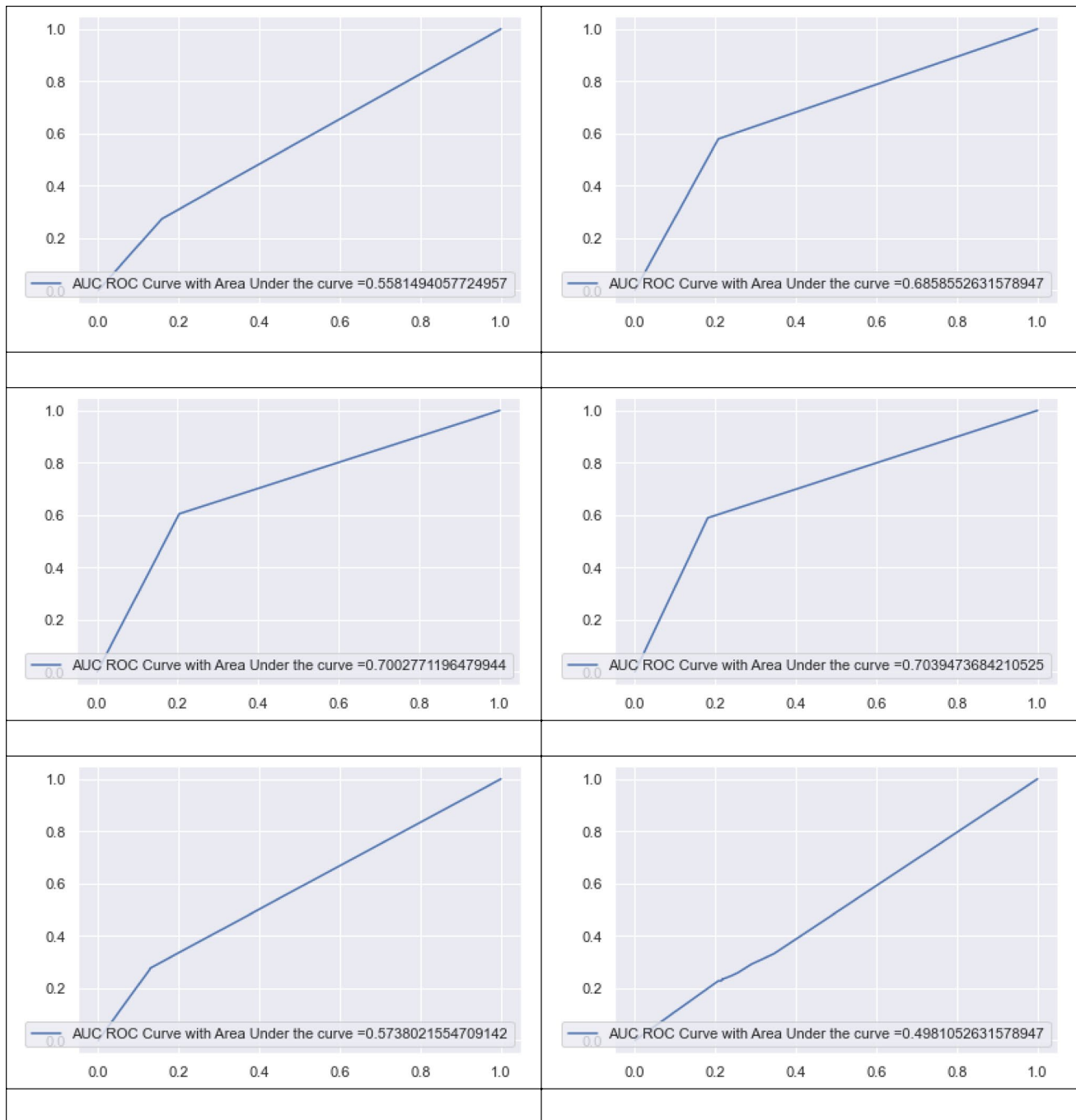


Fig. 8 AUC ROC curves of the BOML algorithm implemented on the FHS dataset

Discussion

Table 2 clearly depicts the high imbalance in the FHS dataset when compared with the Statlog dataset. These two datasets are taken into consideration to analyze the behavior of the BOML model on highly imbalanced and lightly imbalanced datasets.

The performance metrics—accuracy, f1score, precision, and recall—are tabulated for the datasets subjected to the CART decision tree classifier alone and presented in Table 4. The same metrics are exhaustively calculated for the datasets subjected to the BOML algorithm, and the results are tabulated in Table 3. The comparison between the values in Table 3 and Table 4 depicts significant improvement in

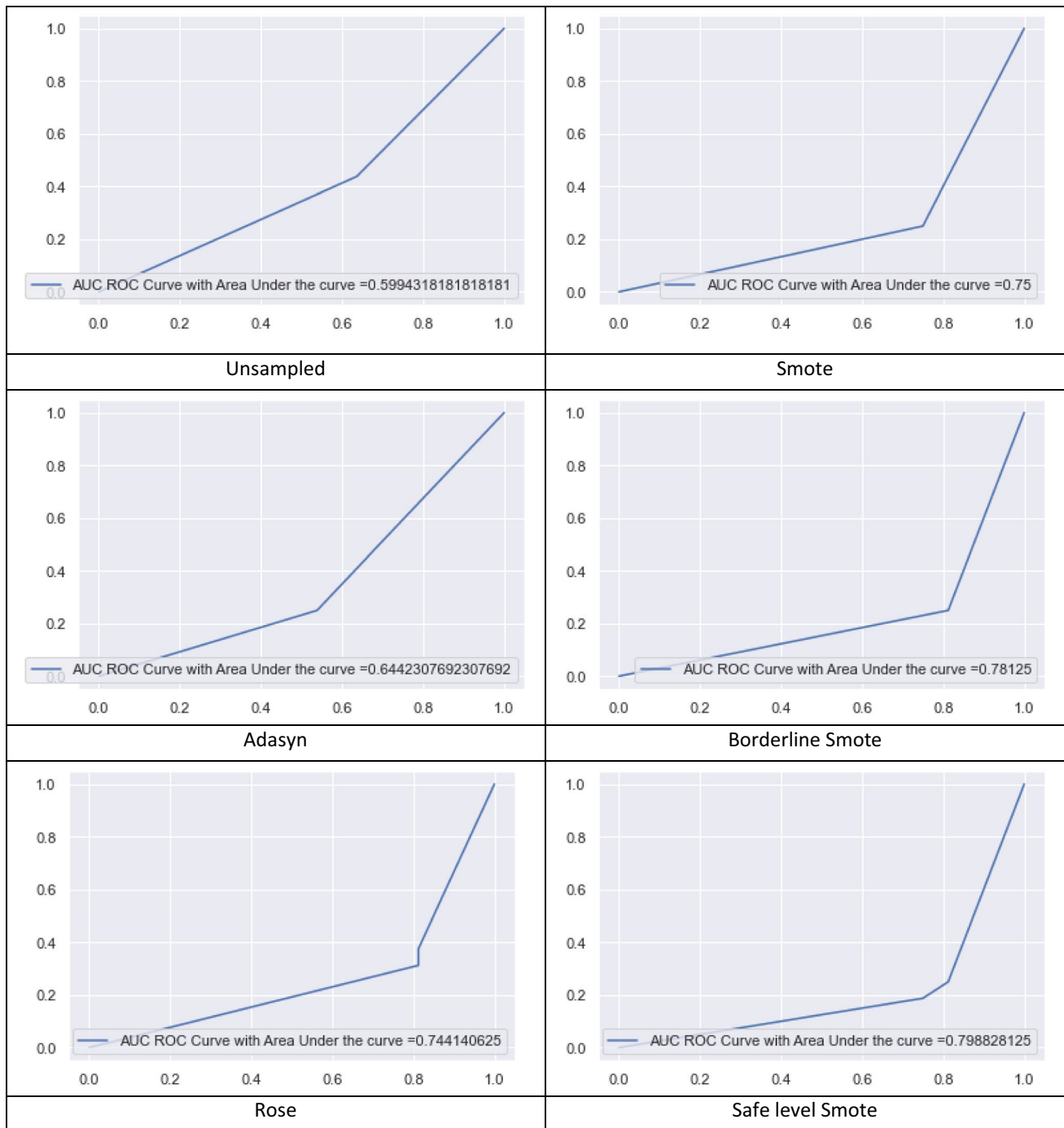


Fig. 9 AUC ROC curves of BOML algorithm implemented on Statlog dataset

performance metrics, when the datasets are subjected to the BOML algorithm.

The datasets under consideration focus on the prediction of heart disease. Consequently, though accuracy is important, the focus should be on reducing false negatives as human lives cannot be put at risk. A higher value

of recall and f1 score concentrates on reduced false negatives. Analysis of values in Table 3 shows that for the FHS dataset, SMOTE sampled dataset implemented on the BOML algorithm gives the highest performance when compared with the unsampled FHS dataset, with an improvement of 11%, 75%, 62%, and 71% in accuracy,

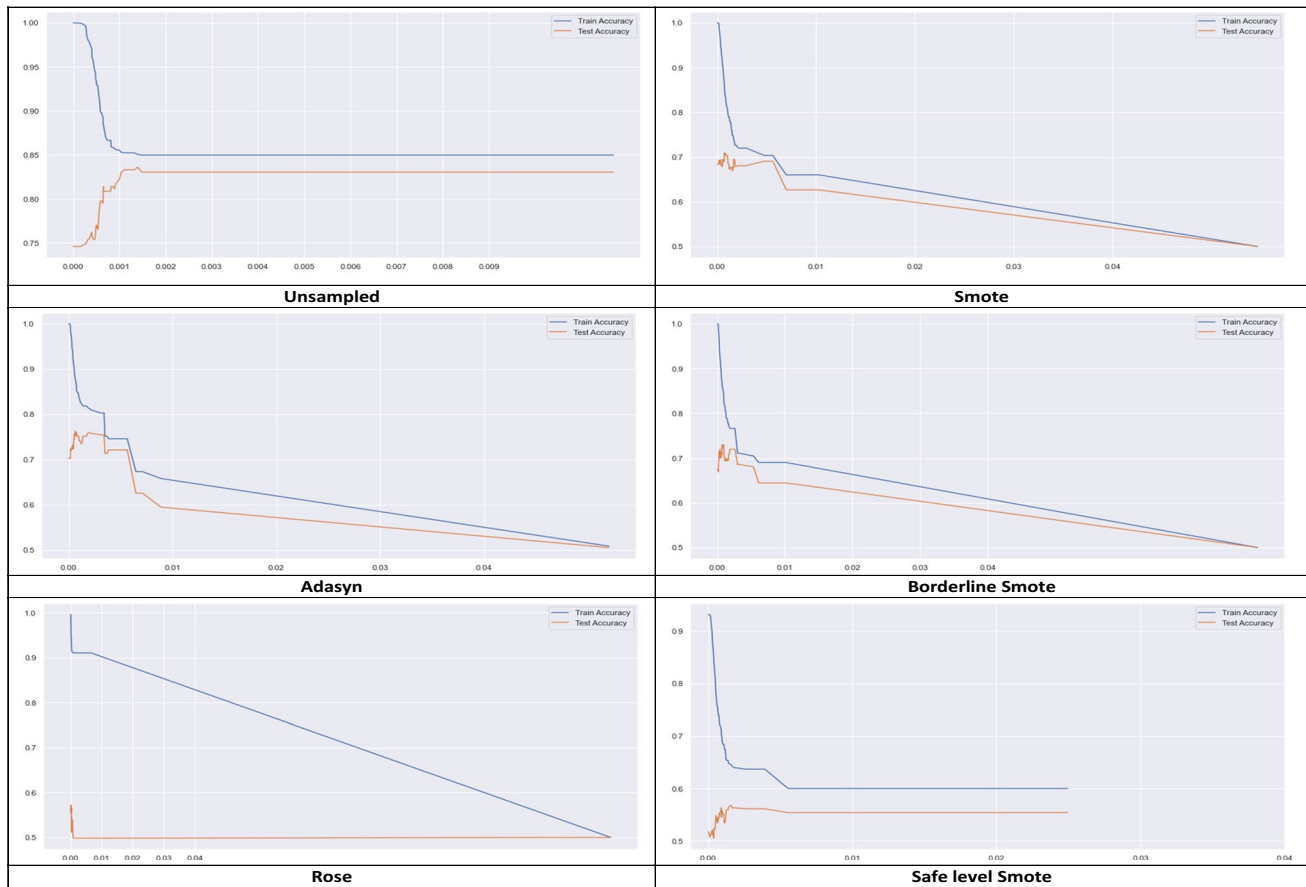


Fig. 10 Training and testing accuracies for various values of cost complexity alpha parameter—FHS dataset

precision, recall, and f1 scores. For the Statlog dataset, Adasyn sampled dataset implemented on the BOML algorithm gives the highest performance when compared with the unsampled Statlog dataset, with an improvement of 17%, 27%, 25%, and 05% in accuracy, precision, recall, and f1 scores. The results additionally signify better performance of the BOML algorithm on the FHS-highly imbalanced dataset when compared with the Statlog-lightly imbalanced dataset.

Since the BOML algorithm deploys oversampled datasets, it should be ensured that the training and test accuracies do not deviate significantly. Figures 10 and 11 portray that

there is no significant difference in the training and testing accuracies except for the rose-sampled FHS dataset. This ensures that the BOML model is not overfitting and will behave well for unseen data.

Attribute pruning was exhaustively done for each case. The results depict that the significant attributes that contribute to a higher risk of heart disease in the FHS dataset are age, heart rate, blood pressure, glucose, BMI, and cholesterol. For the Statlog dataset, the key attributes that contribute to the presence of heart disease relate to the major vessels identified by fluoroscopy, age, and chest pain type. These factors contribute to the prevention of disease.

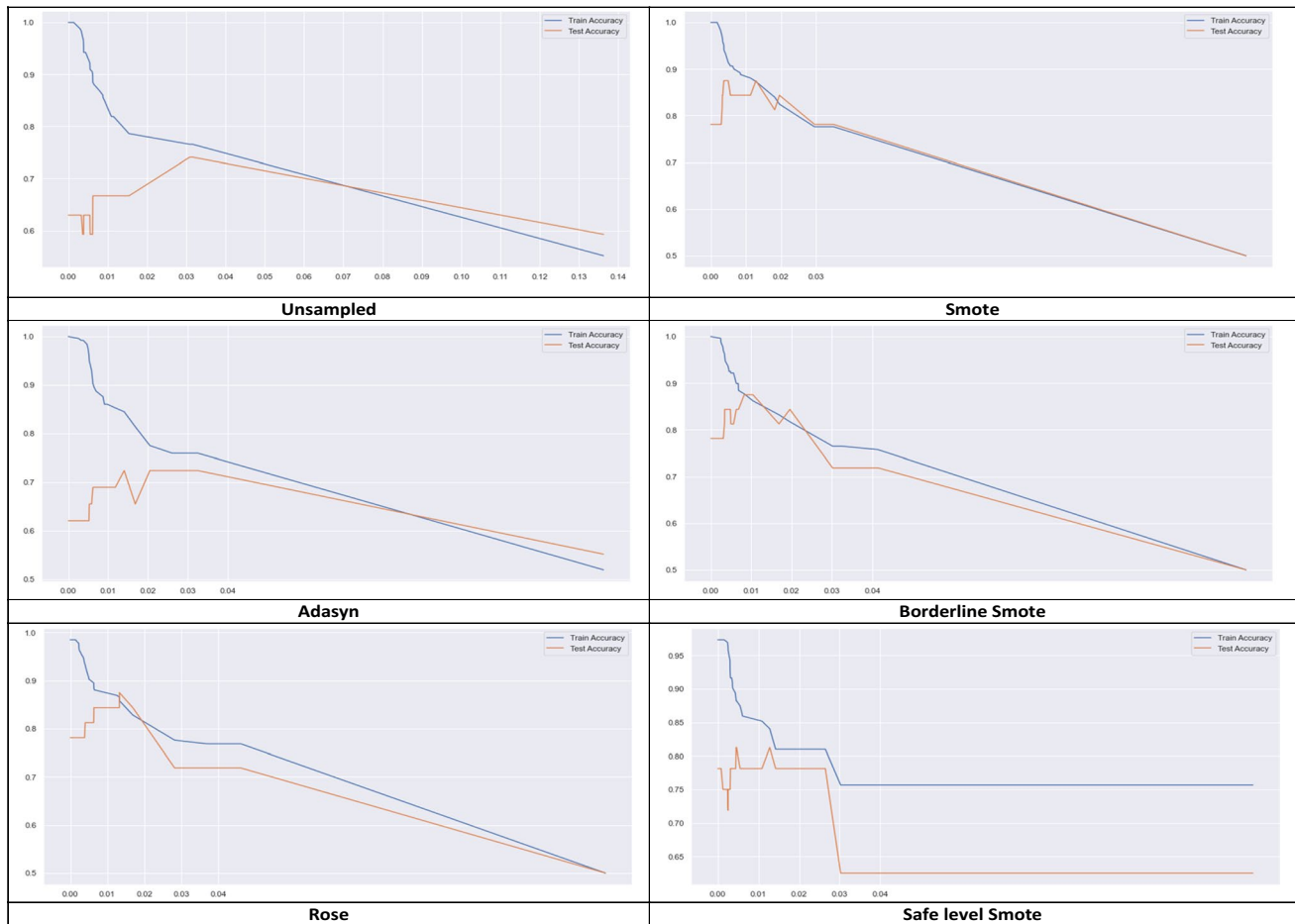
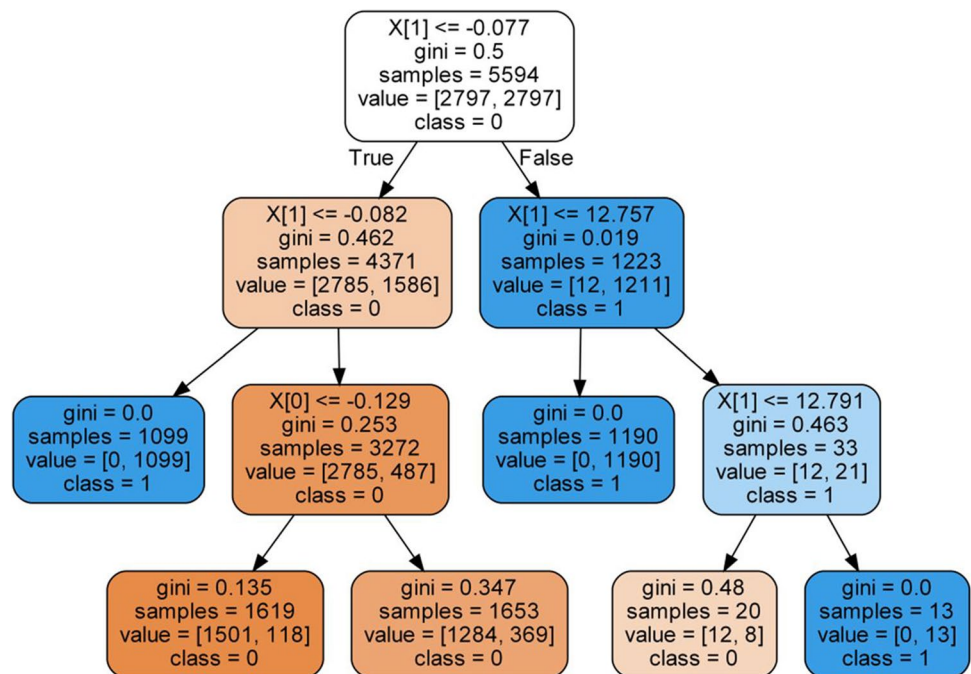


Fig. 11 Training and testing accuracies for various values of cost complexity alpha parameter—Statlog dataset

Fig. 12 Decision tree generated for BOML algorithm implemented on rose sampled FHS dataset



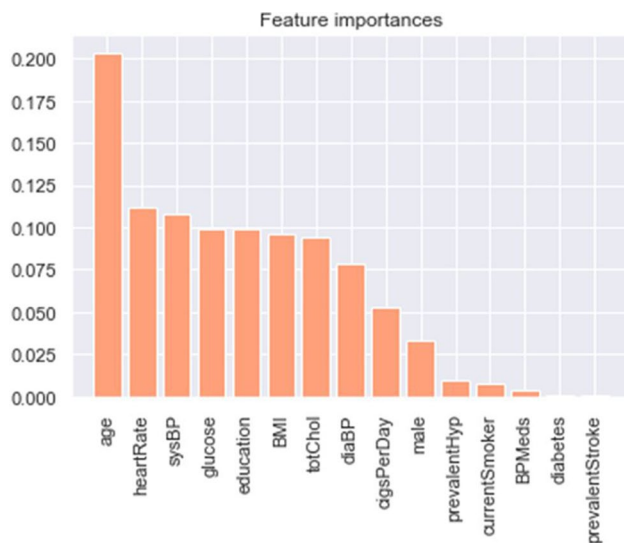


Fig. 13 Attribute scores for SMOTE sampled FHS dataset

Conclusion

The proposed BOML method, a novel algorithm for highly imbalanced datasets, exhibits a hybrid approach of oversampling, CART decision classifier with attribute and rule pruning for effective detection of heart disease. The novelty of the algorithm is the unique hybrid approach that works effectively on imbalanced datasets. Experimental results conclude SMOTE sampling-based BOML model is highly effective for the highly imbalanced FHS dataset and the Adasyn sampling-based BOML model is effective for the less imbalanced Statlog dataset. The algorithm has effectively identified the key attributes that impose a higher risk of heart disease. Monitoring and controlling factors such as age, heart rate, blood pressure, glucose, BMI, cholesterol, blood vessels, and chest pain type will contribute to the prevention of heart disease. However, future work focuses on the validation of the results by clinicians. Futuristically, the work will focus on a unified methodology that will automatically choose the right sampling and pruning strategies.

Data availability The dataset analyzed during the current study is the UCI machine learning repository dataset which is a publicly available dataset, and it is downloaded from the URL [https://archive.ics.uci.edu/ml/datasets/statlog+\(heart\)](https://archive.ics.uci.edu/ml/datasets/statlog+(heart)).

Code availability The developed code of this study will be shared upon request.

Declarations

Ethical approval This article does not contain any studies with human participants performed by any of the authors.

Informed consent There is no individual participant included in the study.

Conflict of interest The authors declare no competing interests.

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